Advanced Technologies for Biodiversity Monitoring:

A Data Pipeline for Flower Detection, Counting and Tracking

Authors

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01 Introduction

Biodiversity monitoring has a crucial role in promoting sustainable coexistence between humans, nature, especially in sight of the EU Green Deal [1].
Biodiversity is a marker of ecosystem stability. It necessitates the effective collection and swift analysis of data, over extended periods (entire seasons) and large scale program.

On the other hand, its **continuous evaluation** is a complex, laborious manual task in which dedicated biodiversity personnel wastes their scarce operational resources.

The richness of flowers has strongly deteriorated [2]. It perpetuates a cascading **multiplier effect at different levels**, e.g., impacting insects which in turn on animals and quality of life.

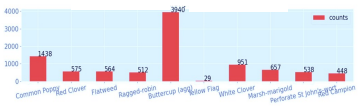
02 Objectives

The objective of the project is **monitoring the distribution of flowers** over large stretches around agricultural areas.

Data Collection in rural area.

Developing a **proof-of-concept to extract, identify and count flowers in photos**.

Generating an annotated flower dataset and develop a data tool to extract and count flowers in photos.



Distibution of Dataset (1001 photos) with annotations of 10 species of flowers

03 Methodology

In response to these limitations, our proposed experimentation features:

1) **The utilisation of drones and professional cameras** for Image Acquisition in rural areas

2) The subsequent **development and implementation** of Image Analysis techniques– applied state-of-the-art YOLO (You Only Look Once [2]) algorithms

Our study features a dataset which encompasses a collection of images of flowers in geofenced rural areas, collected during the Dutch spring and summer.

Data Collection in Dutch rural field

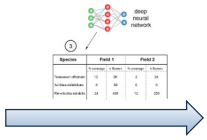


Image detection

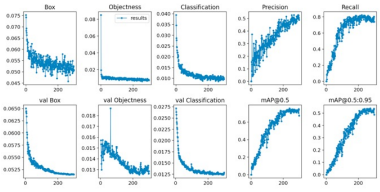
Development of User Interface (UI) designed for single photo or folders

04 Analysis/Results

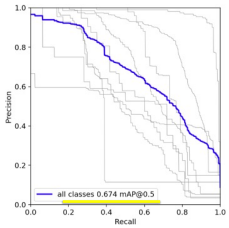
We achieved the objectives, **generating high-quality annotations** for the selected species with 18443 bounding boxes, trained YOLOv4 and YOLOv5 models, developed Web Applications as a user-friendly access to trained model.

	Model	Transfer Learning	Img size	Batch	Epoch	Mosaic	GPU	OS
A	YOLOv4	False	640	2	300	True	NVIDIA Quadro T1000	Ubuntu20.04
B	YOLOv4	True	640	2	300	True		
C	YOLOv4	True	960	2	300	True	NIVDIA Tesla M60	CentOS Linux Release 7.9.2009(Core)
D	Modified YOLOv4[12]	False	640	2	300	True		

Training parameters of 4 Models (A,B,C and D) *in the development phase.*
Randomly the dataset was splitted (1001 photos) into ratio 0.7/0.2/0.1 for train, validation and test sets.



Training Result of Model C *with the highest value of the mean average precision (mAP)*



Performance of Model C

*The **mean average precision** (mAP) is the value for **Object Detection models** (like R-CNN and YOLO).

Despite the feeling that humans are masters of the world, even the tiniest and invisible entities have a voice and wield power to govern it!"

05 Conclusion/Future works

We generated high-quality **Annotated Flower Datasets** for biodiversity and computer vision communities.

The choice of method (Object Detection instead of Image Classification) turned out a strategical success. The tool helps and perform in the **auto-annotation** process. It is faster respect human.

The future goals are improving the model's performance, **experimenting with YOLO family** and parameters and **number of species to identify**, **increasing the dataset** with recent images collected.

Publication of the work, **sharing the datasets** and finding collaborations.

Related Literature and links

- [1]. **EU Biodiversity Strategy for 2030: Bringing nature back into our lives (2020/2273(INI) :** https://eur-lex.europa.eu/legal-content/EN/TXT/HTML/?uri=CELEX%3A52021IP0277
- [2] IPBES-IPCC Workshop Biodiversity and climate change (2021).
- https://www.floravannederland.nl/
- **Annotation Tool:** https://github.com/xiaohenghuang/labelImgGo
- **CICD result Github repository:** Github repository: https://github.com/xiaohenghuang/cdcheck
- Datasets sharing soon or contact me by email

SCAN ME

Article on IoT Beehive for Urban Biodiversity